The shallow structure of Kīlauea caldera from high-resolution Bouguer gravity and total magnetic anomaly mapping: Insights into progressive magma reservoir growth

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Received 17 October 2012; revised 30 May 2013; accepted 7 June 2013; published 17 July 2013.

[1] We conducted total magnetic field and Bouguer gravity measurements to investigate the shallow structure beneath the summit caldera of Kīlauea Volcano, Hawai'i. Two significant and distinctive magnetic anomalies were identified within the caldera. One is interpreted to be associated with a long-lived prehistoric eruptive center, the Observatory vent, located ~1 km east of the Hawaiian Volcano Observatory. The second magnetic anomaly corresponds to a set of eruptive fissures that strike northeast from Halema'uma'u Crater, suggesting this is an important transport pathway for magma. The Bouguer gravity data were inverted to produce 3-D models of density contrasts in the upper 2 km beneath Kīlauea. The models detect 3.0 km^3 of material, denser than 2800 $kg\,m^{-3}$, beneath the caldera that may represent an intrusive complex centered northeast of Halema'uma'u. Recent temporal gravity studies indicate continual addition of mass beneath the caldera during 1975–2008 centered west of Halema'uma'u and suggest this is due to filling of void space. The growth of a large intrusive complex, apparent cyclical caldera formation, and continual mass addition without inflation, however, can also be explained by extensional rifting caused by the continual southward movement of Kīlauea's unstable south flank.

Citation: Zurek, J., and G. Williams-Jones (2013), The shallow structure of Kīlauea caldera from high-resolution Bouguer gravity and total magnetic anomaly mapping: Insights into progressive magma reservoir growth, *J. Geophys. Res. Solid Earth*, *118*, 3742–3752, doi:10.1002/jgrb.50243.

1. Introduction

[2] Volcanoes are structurally complex due to the processes of intrusion, eruption, and tectonism. Understanding a volcano's structure is critical to hazard monitoring and assessment, as it is a determining factor governing the locations of potential eruptions as well as the type of activity that may occur. This is particularly apparent at Kīlauea Volcano, Hawai'i, which is characterized by effusive and explosive eruptions in the summit caldera and along its rift zones, as well as flank instability that may promote development of the shallow magmatic system through rifting [e.g., *Delaney et al.*, 1998].

[3] To further investigate the shallow magmatic system of Kīlauea, high-resolution Bouguer gravity and total magnetic surveys were completed in April–May 2009 to image subsurface density and magnetic structures. Inversion of Bouguer gravity data provides insight into density contrasts and large-scale structures without relying on prior assumptions of source geometry or substrate homogeneity. Likewise, total magnetic field mapping can identify shallow geologic structures with no surface expression or density contrast. These data also allow for evaluation of models based on previous dynamic gravity [*Johnson et al.*, 2010], deformation [e.g., *Cervelli and Miklius*, 2003; *Montgomery-Brown et al.*, 2010], and seismic studies [*Ohminato et al.*, 1998; *Dawson et al.*, 1999; *Battaglia et al.*, 2003] that infer the presence of magma reservoirs at shallow levels beneath Kīlauea's summit caldera. The association between growth of the shallow magmatic system, caldera formation, and continual mass addition without inflation is investigated here. Finally, we propose a mechanism for the growth of the shallow magmatic system at Kilauea.

2. Geologic Setting and Previous Work

[4] Kīlauea is one of five volcanic edifices that make up the Island of Hawai'i (Figure 1a inset), at the leading edge of a hotspot trend in the middle of the Pacific Ocean. Currently, Kīlauea is the only actively erupting volcano on the island, and it has been in a nearly continual state of eruption since 1983—the Pu'u'Ō'ō eruption began on 3 January 1983 on the volcano's east rift zone (Figure 1a) [*Heliker and Mattox*, 2003]. In addition, an eruption at the summit, which had not experienced eruptive activity since 1982, began in March 2008 with the opening of a vent along the eastern margin of Halema'uma'u Crater; this eruption continues to the present. The summit eruption is characterized by low-level,

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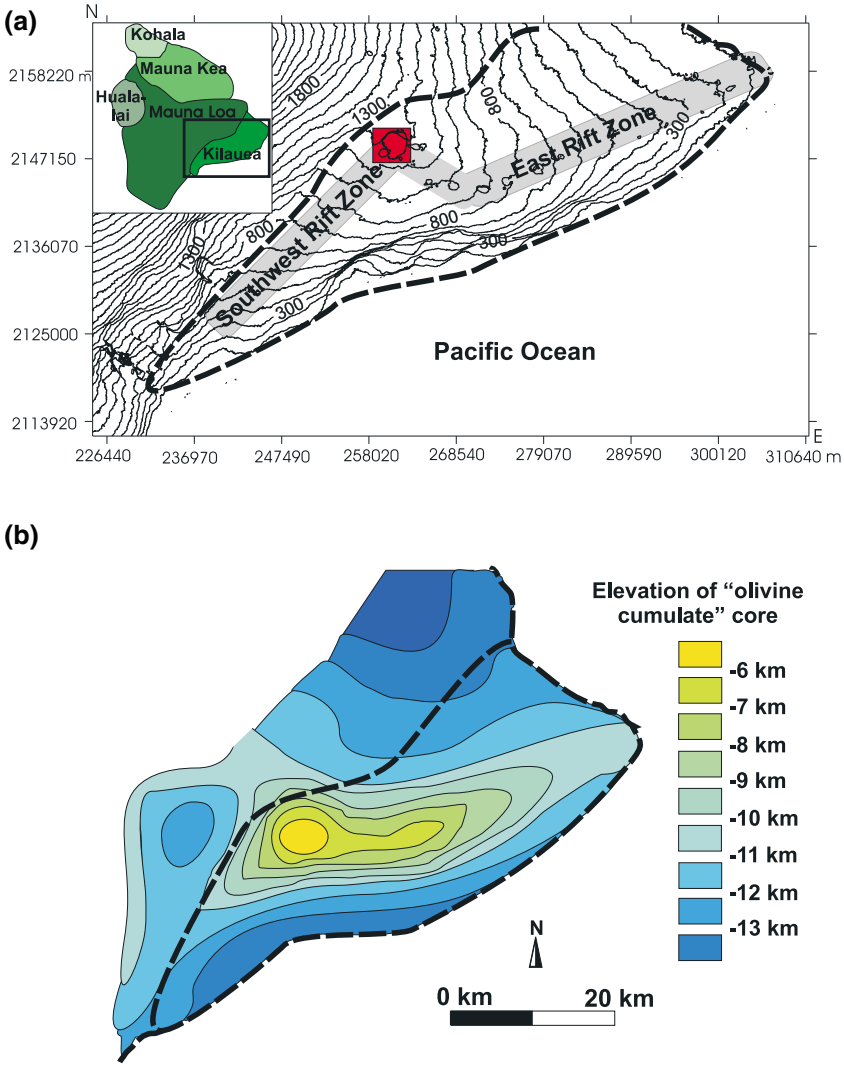

Figure 1. (a) Topographic map of Kīlauea (volcano area is outlined by the dashed line), with the summit caldera and survey area highlighted in red and both rift zones in gray. Inset: Island of Hawai'i and the five volcanic centers that make up the island. (b) Results of a regional Bouguer gravity survey showing the depth to an inferred dense olivine cumulate core (3300 kg m^{-3}) (modified after *Kauahikaua et al*. [2000]).

persistent emission of ash and gas, as well as a lava lake that experiences rise and fall cycles and that is occasionally disrupted by rock falls from the vent rim and walls [e.g., *Wooten et al*., 2009; *Patrick et al*., 2011; *Orr et al*., 2013]. During our study (April–May 2009), the volcano was experiencing nearly steady state eruptive activity from both the summit and Pu'u'Ō'ō. A transient deformation event did occur during the magnetic survey which briefly reduced lava extrusion at the eruption site on the east rift zone. Such events are relatively common at Kīlauea [e.g., *Cervelli and Miklius*, 2003] and small in scale compared to dike intrusions and fissure eruptions [e.g., *Montgomery-Brown et al*., 2010].

[5] The summit of Kīlauea has had a complex history of caldera formation and filling. An older caldera has been inferred to have existed between 1500 and 2100 years ago [*Powers*, 1948]. The current caldera formed about 1470–1510 Common Era based on ^{14}C dating of postcaldera tephra deposits and precaldera lava flows [*Swanson et al*., 2012] and consistent with Hawaiian oral traditions [*Swanson*, 2008]. Since 1790, there has been a net rise in the level of the caldera floor due to resurfacing by lava, leading to the southern end of the caldera being nearly filled [*Holcomb*, 1987]. Geologic mapping of the caldera [*Neal and Lockwood*, 2003] shows the repaving of the caldera with the majority of the surface younger than 100 years (Figure 2). Important structures from Kīlauea's past are likely buried, including old lava lakes and eruptive vents (e.g., the Observatory vent [*Holcomb*, 1987]).

[6] Gravity surveys have been utilized on the Island of Hawai'i to map areas of high density and provide limits on mass flux at Kīlauea. The most recent island-wide Bouguer gravity survey confirmed that the core of each volcano consists of material approaching the density of an olivine cumulate (3300 kg m^{-3}) [*Kauahikaua et al*., 2000]. Through modeling and anomaly wavelength analysis, the depths to the dense core material were calculated for the entire island, with that beneath Kīlauea's summit inferred at 5 to 6 km below the surface and becoming deeper away from the summit

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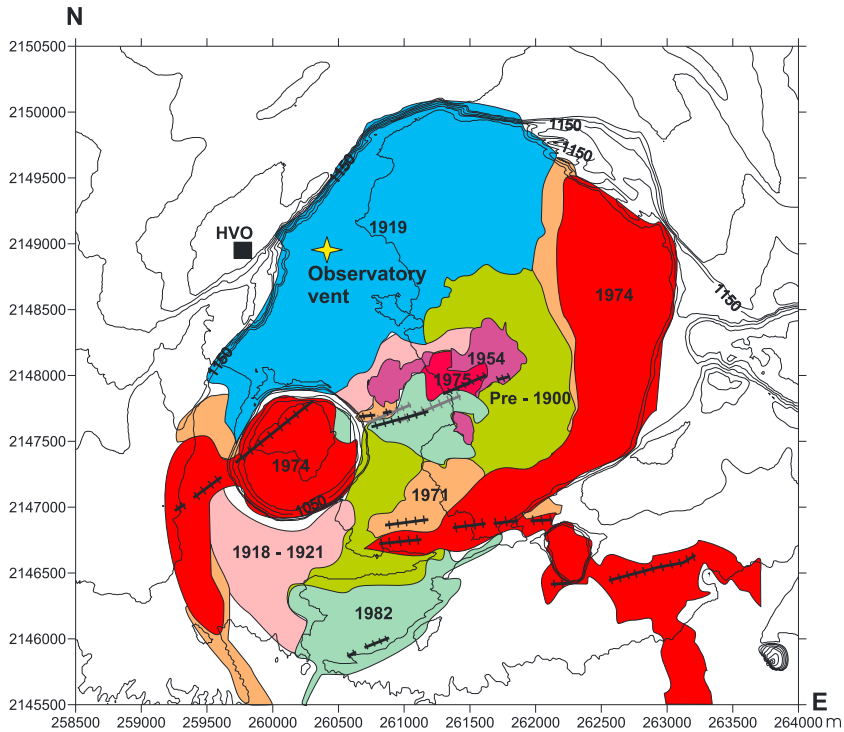

Figure 2. Geologic map of Kīlauea's summit caldera with visible eruptive fissures represented as black lines and buried fissures shown in gray. The inferred location of the Observatory vent from *Holcomb* [1987] is represented by a yellow star and the Hawaiian Volcano Observatory (HVO) by a black square. Modified from *Holcomb* [1980] and *Neal and Lockwood* [2003].

(Figure 1b). This is supported by the presence of high velocity zones detected using 3-D active and passive source tomography [*Park et al*., 2009]. The Bouguer anomaly study also indicates that dense material underlies Kīlauea's rift zones. While the spatial coverage of this survey provided excellent constraints on regional gravity anomalies, it was not able to detect density contrasts that might exist at shallow levels beneath Kīlauea Caldera.

[7] To investigate mass flux within Kīlauea's magma plumbing system, dynamic gravity surveys have been performed across a network of stations in Kīlauea's summit region [e.g., *Kauahikaua and Miklius*, 2003; *Johnson et al*., 2010]. Surveys prior to and following a M7.2 earthquake in 1975 showed a significant decrease in mass beneath the summit, which was interpreted to indicate the creation of $40–90 \times 10^6\,m^3$ of void space due to draining of the magma reservoir and the creation of cracks in the summit region [*Dzurisin et al*., 1980]. Subsequent surveys measured an increasing gravitational field (after correcting for vertical deformation) centered near Halema'uma'u Crater, with a maximum magnitude of approximately 450 μGal over 33 years during a period of net subsidence (~1.9 m)—requiring a complex source mechanism, as the gravity data indicated a mass increase in the subsurface [*Johnson et al*., 2010]. Mechanisms that were discussed by *Johnson et al*. [2010] included olivine cumulates replacing magma, upward migration of the magma chamber, and the filling of void space by magma. Due to the lack of uplift during the 33 year time period, the proposed mechanism for the gravity increase was filling of $21–120 \times 10^6\,m^3$ of void space, similar to the volume of space inferred to have been created following the 1975 earthquake [*Dzurisin et al*., 1980; *Johnson et al*., 2010].

[8] Two large-scale aeromagnetic surveys were flown at different elevations across the Island of Hawai'i [*Godson et al*., 1981; *Flanigan et al*., 1986] which were combined by *Hildenbrand et al*. [1993] to describe the magnetic anomalies displayed by rift zones on Mauna Loa and Kīlauea. These researchers interpreted the short-wavelength positive anomalies over the rift zones as slowly cooled, unaltered intrusions with hydrothermally altered material on either side but cited the need for drill hole data and higher-resolution magnetic surveys to better understand magnetic sources and local anomalies.

[9] Deformation, seismic, and geochemical studies have identified at least two regions of magma accumulation beneath Kīlauea's summit, with the deeper and larger magma chamber located 2 to 4 km beneath the southern part of the caldera [e.g., *Delaney et al*., 1998; *Pietruszka and Garcia*, 1999; *Cervelli and Miklius*, 2003,*Garcia et al*., 2003]. A shallower magma reservoir has been inferred by seismic [e.g., *Ohminato et al*., 1998; *Dawson et al*., 1999; *Battaglia et al*., 2003] and geodetic studies [e.g., *Cervelli and Miklius*, 2003; *Montgomery-Brown et al*., 2010] just east of Halema'uma'u Crater at a depth of approximately 1 km.

3. Methodology and Results

3.1. Magnetic

[10] The total magnetic field data were collected using an Overhauser procession magnetometer. In total, 420 data

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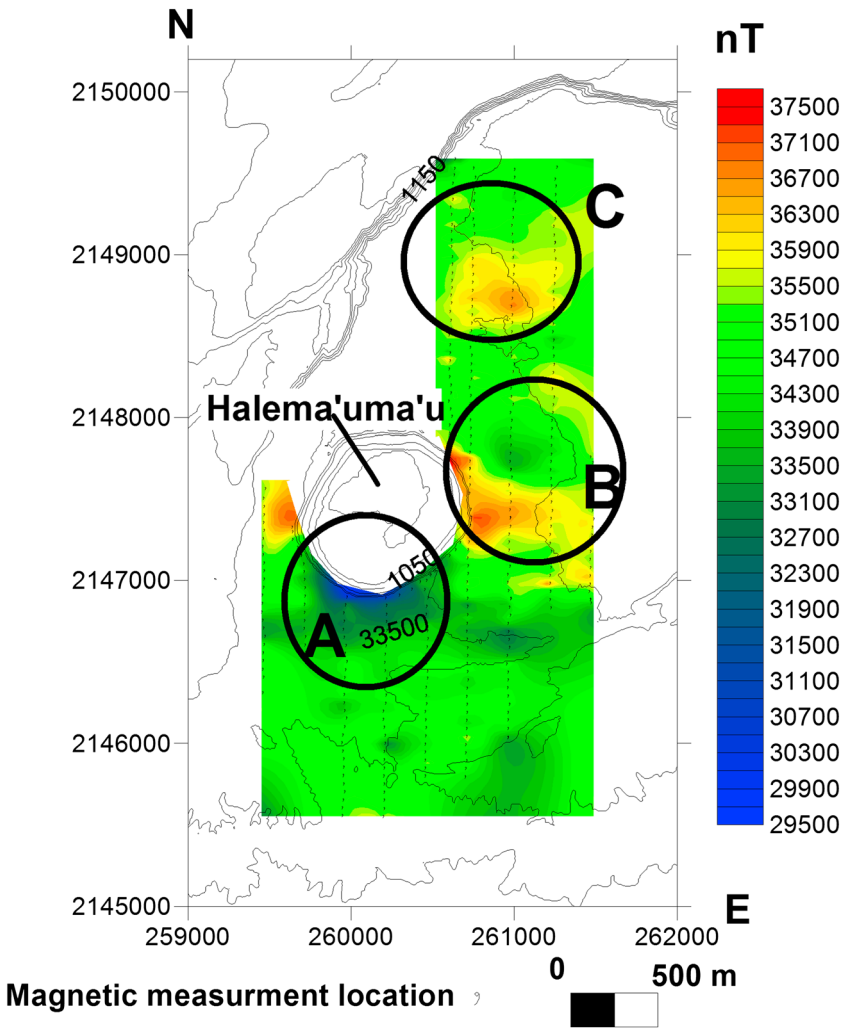

Figure 3. Total magnetic survey map in Kīlauea caldera with three identifiable short-wavelength anomalies outlined by dark circles. The black dots are the measurement locations.

points were taken approximately every 25 m along lines 200 m apart with the sensor mounted on a 2 m surveying pole (survey area approximately 1.6 km by 3.2 km; Figure 3). Although the survey area is sufficient to image magnetic structures to ~1000 m, it is insufficient to filter out the effect of the Curie depth. The depth of the Curie point changes across the caldera, requiring a larger survey area to apply filters without aliasing the data. Therefore, a depth of 500 m is considered the maximum depth for the data set. Measurement locations were obtained from a handheld GPS, with an accuracy of ~5 m, every 5 s while surveying. The time stamp from the magnetometer was later compared to that of the GPS to extrapolate a position for each data point. This provided the ability to cover large areas in a single day, albeit with reduced positional accuracy.

[11] Diurnal variation in solar radiation can affect the local magnetic field by approximately 30 nT [e.g., *Telford et al.*, 1990]. To confirm that the daily variation was minor, measurements were taken with a second magnetometer (a GSM-19 W) for 6 h spanning a survey day at a single location approximately 100 m northeast of the Hawaiian Volcano Observatory (HVO; Figure 2). The measured diurnal variation was ~45 nT, which is much smaller than the variations recorded across the caldera. Highly magnetized intrusions and areas of hydrothermal alteration typically produce contrasts greater than 1000 nT [e.g., *Telford et al.*, 1990; *Hildenbrand et al.*, 1993]; therefore, diurnal corrections can be ignored. To eliminate the unlikely possibility that a solar storm might affect the results, repeat measurements were made throughout each survey day to verify the stability of the instrument and surrounding magnetic field (repeatability was better than 2 nT). Magnetic data from a U.S. Geological Survey station in Honolulu were subsequently used to confirm that no magnetic storms took place during surveying. One additional source of error in the final data set stems from not removing the International Geomagnetic Reference Field due to errors in vertical position of each measurement and the lack of significant topography (Text S1 in the supporting information). Taking into consideration all sources of error, the survey has an estimated uncertainty of 150 nT.

[12] The processed and gridded magnetic data show three distinct anomalies that deviate from the background of ~34,500 nT. The largest is a magnetic low, ~4000–5000 nT in magnitude, associated with the southern edge of Halema'uma'u Crater (Figure 3; anomaly 1). Two other small, well-defined anomalies to the east of Halema'uma'u are ~3000 nT in amplitude and no more than 500 m across.

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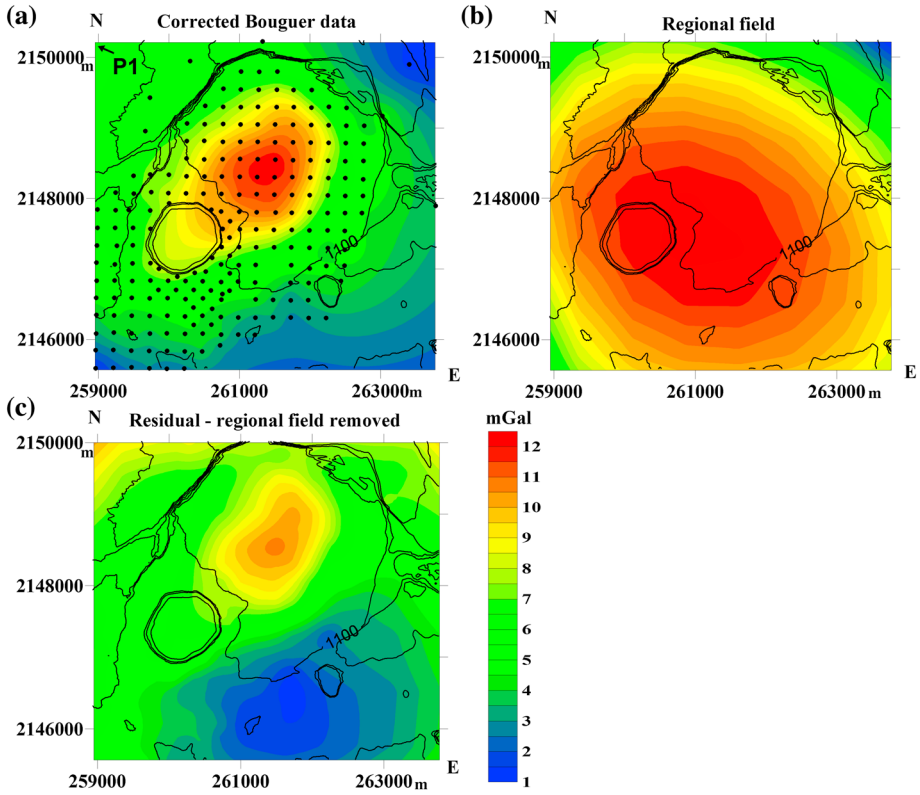

Figure 4. (a) Bouguer gravity anomaly map corrected for terrain, earth tides, and normalized to the base station. Black dots are the measurement locations. P1 is the base station used by each gravity survey and is located just off the corner of the map. (b) The regional field is calculated through a two-step inversion using data from *Kauahikaua et al*. [2000] (c) Residual Bouguer gravity anomaly map where the regional field from part B has been removed.

Many other short-wavelength changes are apparent within the survey area; however, these small-scale anomalies were not analyzed in detail due to the lack of definition in their magnetic structure. To aid in the interpretation of the magnetic data, a model was created to test the effect of topography on the local magnetic field of Kīlauea (Figure S1 and Text S1 in the supporting information) using a 5 m-resolution digital elevation model and the magnetic modeling software package MAG3D [*MAG3D*, 2007].

3.2. Gravity

[13] Gravity measurements were collected at 231 stations using a LaCoste and Romberg gravimeter (G-127) equipped with an Aliod electronic feedback system. Detailed gravity survey techniques have been discussed extensively in the literature [e.g., *Rymer and Brown*, 1986; *Berrino et al*., 1992; *Battaglia et al*., 2008] and will only be reviewed briefly here. The station grid spacing at Kīlauea was 250 by 250 m across the whole caldera except in Halema'uma'u Crater; infill stations were therefore added on the eastern side of the crater to constrain any gravity variations associated with the summit eruptive vent (Figure 4a). Each survey used a base station located at benchmark P1 (Figure 4a) and regular station repeats to identify anomalous instrumental drift (closure) and data tares. The error on each gravity measurement varies due to daily survey closures that were typically less than 100 μGal. Access to many of the areas surveyed was by foot over broken terrain, reducing repeat base station measurements and limiting the ability to identify and correct for tares; this is the most probable cause for the large closure errors [e.g., *Crider et al*., 2008; *Zurek et al*., 2012]. The base station P1, located a few kilometers NW of the caldera, was chosen to reduce seismic noise associated with the summit eruptive vent and for consistency with previous temporal gravity surveys [*Kauahikaua and Miklius*, 2003; *Johnson et al*., 2010]. At the time of our survey, volcanic tremor was occurring at the summit of Kīlauea, increasing noise in gravity readings near the vent by 5 to 20 μGal based on repeat measurements. Normal, laboratory-controlled, daily instrumental drift for G-127 is approximately 10 μGal. Because this level of drift is much smaller than the error associated with the survey, the effect of drift is ignored. Base station measurements were used to normalize the data for each survey day and eliminate instrumental drift and unrecoverable tares that can occur over several weeks.

[14] To obtain the necessary vertical and horizontal control for each gravity station, kinematic GPS surveys were conducted using a continuous GPS station in the summit region as a base and a 1 s sampling interval. Postprocessing provides vertical accuracy on the order of a few centimeters. Warping of the Earth's geoid in the summit region of Kīlauea causes a departure of approximately 22 m from the true elevation above sea level, so the accuracy of station positions is not absolutely constrained. The spatial variation of the geoid within the survey area is planar, however, and less than 1.3 m across the caldera [*Smith and Roman*, 2001]. This is

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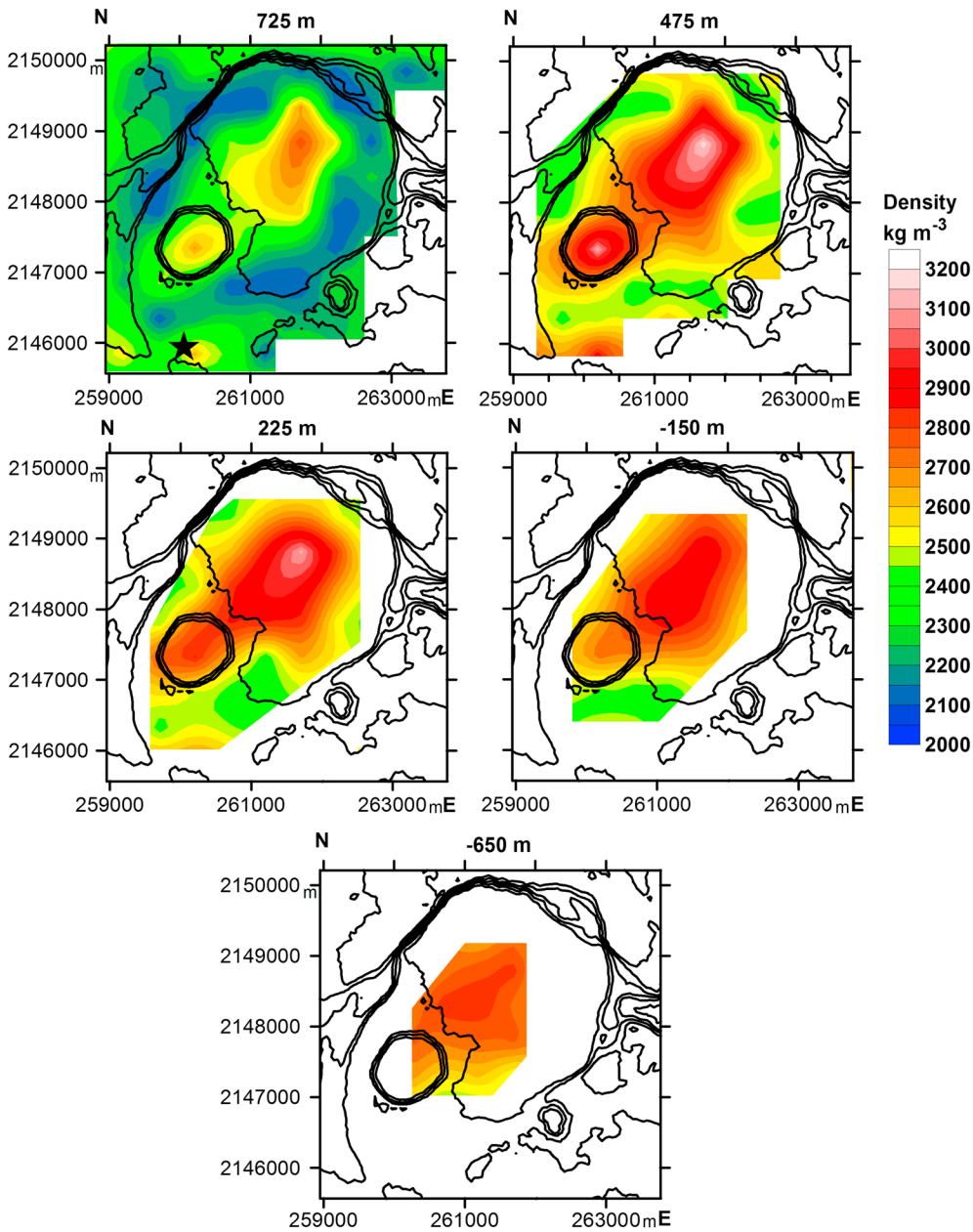

Figure 5. Smoothed depth slices from the Chifact inversion of the gravity data where the density is derived from downhole geophysical data from Keller Well [*Keller et al.*, 1979]. The well—a 1262 m deep borehole located on the caldera rim, 1 km south of Halema'uma'u Crater—is represented by a black star in the 725 m depth slice. All depths are in meters above sea level, and overlain contours are 25 m.

much less than the magnitude of the gravity signal; therefore, the geoid variation can be ignored.

[15] If noise and errors are ignored, the smallest wavelength (Nyquist Frequency) over which a gravity anomaly can be theoretically resolved in this data set without aliasing is 500 m. Due to a lack of further constraints, this theoretical value is used for the forward and inverse models [*Nettleton*, 1940]. The maximum wavelength that can be described is equal to the dimensions of the survey area—approximately 4 km—whereas the maximum and minimum resolvable depth is dependent on both geometry and the maximum wavelength. For an infinite horizontal rod, the depth to the body can be calculated from the anomaly's wavelength such that the depth, Z, is one half of its wavelength, X:

$$X_{1/2} = Z \tag{1}$$

[16] A sphere has a similar equation for determination of the depth to its center where the half-wavelength of the anomaly is multiplied by a factor of 1.3:

$$1.3X_{1/2} = Z \tag{2}$$

[17] If the maximum wavelength is 4 km, then for these simple geometries, the maximum depth our data set can

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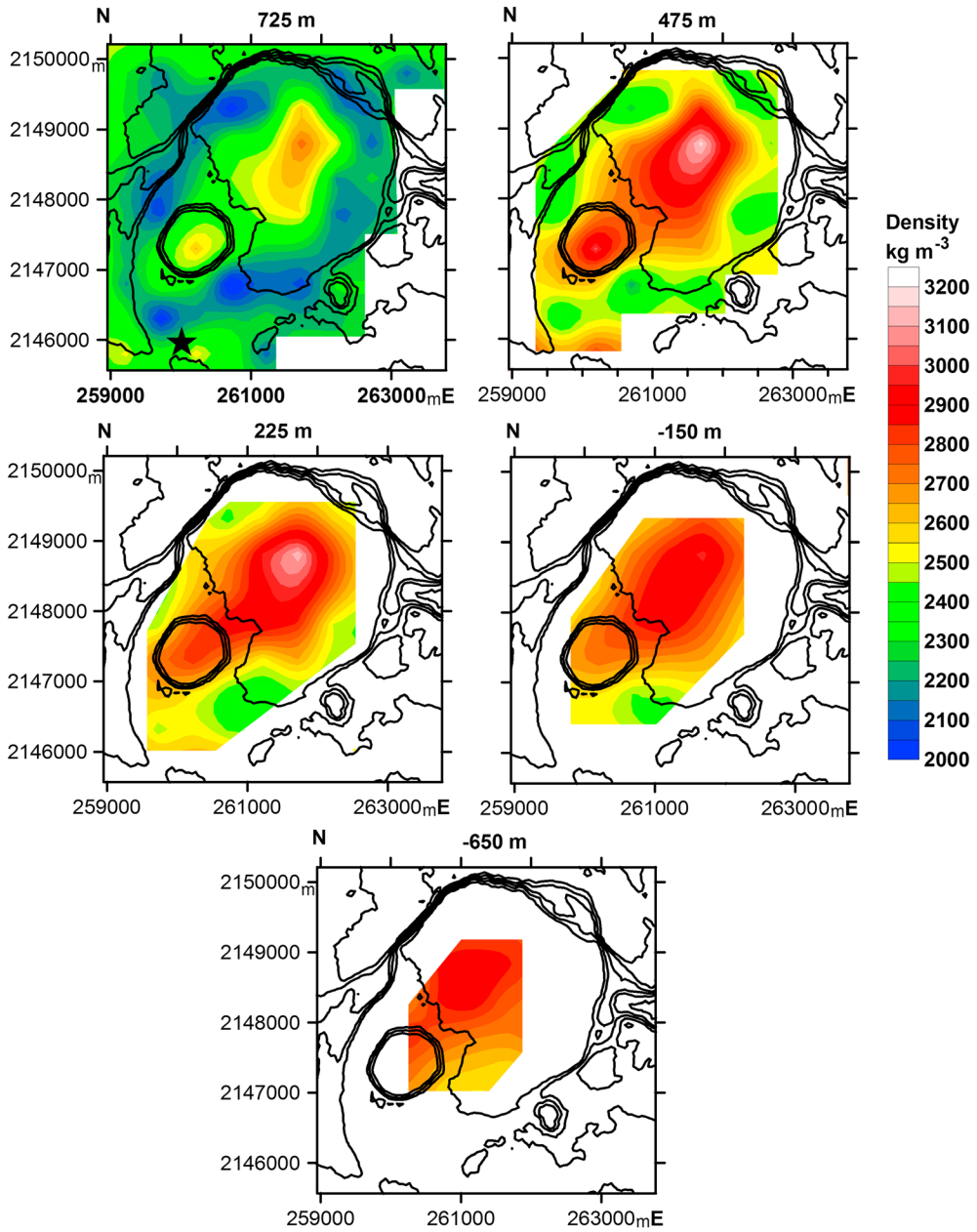

Figure 6. Smoothed depth slices from the GCV inversion of the gravity data where the density is derived from downhole geophysics data from Keller Well [*Keller et al.*, 1979]. Contours and labels are as in Figure 5.

detect is between 2 and 2.6 km, and the minimum is between 250 m and 325 m.

[18] Once corrected for terrain (using an average density of 2300 kg m^{-3} based on previous gravity studies) [*Kauahikaua et al.*, 2000] and free-air effects, our Bouguer anomaly map (Figure 4a) has the same relative magnitude (11 mGal) and shape in the summit area of Kīlauea as that of *Kauahikaua et al.* [2000]. These data sets, however, contain contributions from both regional and local density anomalies. To obtain a residual Bouguer anomaly map highlighting only local density variations requires removal of the regional gravitational field [*Nettleton*, 1940]. There are many different techniques to accomplish this, including fitting a surface to adense basement or taking the second derivative of the data to enhance near-surface effects [e.g., *Gupta and Ramani*, 1982]. In an attempt to reduce the loss of wavelength information due to over processing, we removed the regional gravitational field using a two-step inversion process. First, the regional data set of *Kauahikaua et al.* [2000] was inverted using GRAV3D [*GRAV3D*, 2007] to produce a 3-D density model of Kīlauea and the lower slopes of Mauna Loa. Next, an area 500 m wider, longer, and deeper (3000 m) than our survey area was set to a zero density contrast, and the regional density model created in the first step was forward modeled to obtain the gravity effects of all areas except the shallow area beneath the caldera (Figure 4b). This new regional gravitational potential field was then subtracted from the original data to give the residual Bouguer anomaly

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(Figure 4c). The effectiveness of the two-step inversion process was tested by empirically fitting a polynomial to the data and subtracting that function from the regional field. The remaining signal was nearly identical to the residual from the two-step inversion.

[19] Due to the lack of constraints with respect to the density structure beneath the summit, two general models of subsurface structure were created based on different inversion methods. The resulting shapes of the density structures at depth from each inversion are similar. The first inversion used misfit curves (Chifact) [e.g., *Lines and Treitel*, 2006] with an assumed error of 500 μGal for each point. This model shows an anomalous density contrast that begins approximately 800 m a.s.l. (above sea level), or at a depth of ~400 m below the surface (Figure 5; inversion misfit Figure S2 in the supporting information). It also shows lower density material that roughly follows the caldera rim. The body reaches its maximum horizontal area at ~350 m a.s.l. and disappears below ~ −900 m a.s.l. The generalized cross-validation (GCV) technique was chosen for the second inversion due to its effectiveness with data sets that have good spatial coverage and positive anomalies [*Haber and Oldenburg*, 2000]. While the resulting inversion displays different sizes for density contrasts beneath the summit of Kīlauea relative to the first inversion method, the same general characteristics are apparent (Figure 6; inversion misfit Figure S2). The GCV model shows the density contrast at approximately the same depth (350 m); however, the body is larger and denser.

[20] To outline possible structures based on their density, the gravity inversion must be converted from density contrast to an absolute density. This was done using downhole density data from the Keller Well [*Keller et al.*, 1979]—a 1262 m-deep borehole located 1 km south of Halema'uma'u Crater (Figure S3). To keep the process as simple as possible, three separate levels were chosen based on the average density profiles from the well. The first level (surface down to 600 m a.s.l.) represents vesicular basalt with a density of 2300 kg m^{-3}, and the second starts at the approximate water table depth (600 m) where water fills the pore spaces of basalt and changes the density to 2600 kg m^{-3}. The average density structure taken from the Keller Well is assumed to be equal to the inversion model's 0 kg m^{-3} density contrast and was thus added to each cell to produce models of "absolute" density. Volume estimates of the dense body beneath Kīlauea's summit caldera were made by calculating the area enclosed by the 2800 kg m^{-3} density contour within each model. This density contour was chosen because solidified dyke densities are typically between 2800 and 3100 kg m^{-3} [*Moore*, 2001]. Both models show almost identical volumes of 3.0 km^3; however, the GCV inversion model is slightly larger (Figure 6).

[21] A sensitivity analysis was completed to assess whether or not our survey could identify void space beneath Kīlauea's summit—a key element of the model proposed to explain the dynamic gravity changes measured by *Johnson et al.* [2010]. A wide range of different void space geometries were forward modeled in GRAV3D to determine anomaly amplitudes versus void volumes. The conclusion from this analysis is that, due to the nonuniqueness of potential fields, it is possible in most cases to reproduce the effect of void space with a lower density rock. Only large voids (10^5 m^3 or higher) can realistically be detected without prior constraints on the geometry at depth. For example, a spherical void with a volume of 10^6 m^3 at a depth of 300 m (approximate minimum depth required to be detected just east of Halema'uma'u Crater in this study) has maximum amplitude of −240 μGal. That void space was then modeled as underlain by a dense olivine cumulate (+700 kg m^{-3}) layer at 650 m depth. The combined effect resulted in a smaller negative anomaly at the surface which could be easily represented by a number of different models that do not include void space; thus, it is difficult to unambiguously assess the volume of any void space that might exist beneath the caldera. We note, however, that there are no negative anomalies within the residual Bouguer gravity data collected at Kīlauea's summit (Figure 4c); thus, any significant void space beneath Kīlauea's summit caldera must be masked by nearby high-density bodies such as olivine cumulates and dense intrusions.

4. Discussion

[22] The forward and inverse models developed to interpret both gravity and magnetic data cannot provide unique solutions. Instead, they provide insights into possible structural configurations at depth. When combined with other geologic and geophysical evidence, however, potential field data provide useful constraints on subsurface properties.

4.1. Magnetics

[23] The total magnetic data show three anomalies within the survey area (Figure 3). Anomaly 1 is a broad low located on the southern edge of Halema'uma'u Crater. The magnetic model created to show the theoretical effect of topography on the data also shows an anomalous low on the southern edge of Halema'uma'u, as well as highs on the east and west side of the crater (Figure S1 in the supporting information). This is consistent with the effects of topography at low magnetic latitudes and suggests that anomaly 1 is entirely due to topographic effects. Anomaly 2, however, extends farther from Halema'uma'u than the change predicted by topography alone. The minimum and maximum values of anomaly 2 are situated near fissures that erupted between 1954 and 1982 and strike northeast from Halema'uma'u Crater (Figure 2). The strength of the magnetic anomaly (~3000 nT), its dipole shape, and the lack of evidence to support a complex source geometry (the large magnetic inclination angle around the summit of Kīlauea, 39.9°, means that simple vertical to subvertical bodies will appear as dipoles) suggest a subvertical magnetic source. Given the number of eruptive fissures striking in approximately the same direction, anomaly 2 probably represents an important structure within Kīlauea Caldera, likely related to shallow magma transport along dykes.

[24] Anomaly 3, approximately 2000 nT above the background field strength, is located in the northern part of the caldera away from any visible eruptive fissures or other volcanic features (Figures 2 and 3). This area is covered by lava flows that were erupted in 1919 [*Holcomb*, 1980], suggesting that the source of the anomaly is buried (Figure 2), but the surface in this area is characterized by a thermal anomaly and surface alteration [*Patrick and Witzke*, 2011]. The magnetic anomaly is not a dipole since there is no corresponding low to the north. It is possible that the magnetic low for anomaly 3 could be just outside the survey area to the north, or perhaps the survey coverage is not sufficiently dense to define it. If the corresponding low is outside the survey area,

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the source of the anomaly would need to be buried at sufficient depth (>700 m) to produce a large wavelength magnetic signature. The distribution of thermal anomalies and alteration products within Kīlauea's caldera traces out a circle, concentric to the caldera rim, that has been attributed to buried scarps created prior to the 1800s that still serve as pathways for rising gas [*Macdonald*, 1955; *Fischer et al.*, 1964; *Patrick and Witzke*, 2011]. It is possible that alteration along these buried scarps is responsible for magnetic anomaly 3; however, alteration typically results in a negative magnetic anomaly [*Ade-Hall et al.*, 1971]. Furthermore, the anomaly does not extend to the edge of the survey as would be expected from a continuous scarp. Another possible source is a buried lava lake, as documented in that region by *Ellis* [1825]. A buried lava lake that has a sufficient depth to form layers (similar to an intrusive body) could produce a larger magnetic signature. Previous geologic mapping by *Holcomb* [1980, 1987] of lava flow directions inferred that a long-lived prehistoric eruptive center, called the Observatory vent, was located approximately 1 km east of HVO (Figure 2). The Observatory flows formed a large shield at the summit and created a flow field that stretched down the southwest rift zone to the ocean [*Holcomb*, 1980]. The ages of the Observatory flows are poorly constrained due to limited carbon samples and a lack of detailed mapping away from the summit, but are estimated at between 470 and 625 years B.P. [*Neal and Lockwood*, 2003]. The date of these flows is likely closer to 625 years B.P. as caldera formation has been dated to ~500 years B.P. [*Swanson et al.*, 2012]. Although there is no evidence to confirm whether anomaly 3 is indeed associated with the inferred Observatory vent, a long-lived eruptive site should be associated with a complex magnetic signature and could provide pathways for heat flow and subsequent alteration.

4.2. Bouguer Gravity

[25] The magnetic data provide the ability to describe major features within several hundred meters of the surface beneath Kīlauea Caldera; however, it is not possible with this data set to image deeper due to topography and a shallow Curie point. Gravity data are not affected by or can be corrected for these factors; therefore, by combining magnetic and Bouguer gravity data, we can extend our ability to detect structural changes to a depth of approximately 2 km (Figures 5 and 6). A previous Bouguer gravity survey [*Kauahikaua et al.*, 2000] imaged a dense core to Kīlauea that is much deeper than we can resolve, but that survey was not as sensitive as our study to the upper few kilometers. Combining the results from both surveys suggests that Kīlauea's dense core may start at shallow levels, broadening and becoming denser with depth.

[26] Inversions of our gravity data set suggest that a dense body begins at approximately 800 m a.s.l. and reaches a maximum horizontal extent at 350 m a.s.l. We infer the body to consist primarily of solidified magma and lower density material that roughly occupies the same area as the caldera (Figures 5 and 6). The volume of the dense body beneath the summit is ~3.0 km^3 as calculated from the inversion models presented above. Based on the depth, position, and density range of the imaged body, we suggest that it is most likely an intrusive complex consisting mainly of solid rock with densities based on Keller well density profiles [*Keller et al.*, 1979] between 2600 and 3200 $kg\,m^{-3}$. To obtain an estimate of the amount of intrusive material within the caldera, we use two end-member materials with the densities of basaltic lava flows (2300 $kg\,m^{-3}$ above the water table and 2500 $kg\,m^{-3}$ below it) and solidified dykes (2800 $kg\,m^{-3}$). This simplified binary model suggests that ~20% of the volume of the caldera has a density of 2800 $kg\,m^{-3}$ or greater. Furthermore, in an effort to estimate the volume of olivine cumulates beneath the caldera, a similar end-member calculation using densities of 2800 and 3200 $kg\,m^{-3}$ (for inversion model blocks with densities over 2800 $kg\,m^{-3}$) results in a volume of 0.7 km^3. The lower density material shown in the inversions that surrounds the dense body correlates with major caldera bounding faults, suggesting that the presence of faults and cracks has reduced the density of the basalt surrounding the caldera.

[27] *Swanson et al.* [2012] place the date of Kīlauea's most recent caldera formation event at ~500 years ago and infer that the collapse created a depression ~400 m deeper than the caldera is today. The 3.0 km^3 dense body inferred in this study begins at 350 to 400 m below the surface and is consistent with the level of the postcollapse caldera floor. This suggests that the anomalous dense volume imaged here may incorporate the remnants of older magma chambers present at the time of caldera formation—in other words, portions of the magma reservoir that drained and into which the caldera collapsed. Such a drained magma reservoir probably consisted of dense solidified intrusions and olivine cumulates, which would result in the shallow gravity high (11 mGal) that is seen today (Figure 4c). This high is perched on top of the broader gravity high imaged by *Kauahikaua et al.* [2000] (Figure 1b) and represents the upper portion of a complex structure created by repeated intrusions, caldera collapses, and other processes that have occurred over the life of the volcano.

4.3. Magma Reservoir Growth Implications

[28] Geologic data suggest that two caldera forming events occurred during the last 2200 years [e.g., *Powers*, 1948; *Swanson et al.*, 2012]. If caldera formation is cyclic, the density structure at depth may reflect such repeated processes. This requires linking long-term processes at Kīlauea to both the development of a large (3.0 km^3) dense body and caldera formation. One such process is the slow seaward movement of the south flank of Kīlauea [e.g., *Denlinger and Okubo*, 1995], which causes extension along both rift zones [e.g., *Owen et al.*, 1995; *Delaney et al.*, 1998; *Cayol et al.*, 2000; *Montgomery-Brown et al.*, 2010]. South flank deformation also extends into the summit area, where extension is clear from trilateration and GPS data spanning the caldera [*Delaney et al.*, 1998; *Cayol et al.*, 2000; *Owen et al.*, 2000; *Cervelli and Miklius*, 2003]. Rifting of the summit may also act to accommodate magma storage without an increase in reservoir pressure and without causing surface uplift [*Johnson*, 1992].

[29] The orientation of the 1954–1982 intracaldera eruptive fissures, as well as the associated magnetic anomaly (Figure 3), is consistent with north-south summit rifting. That magma pathway is parallel to Kīlauea's east rift zone, which is also dominated by extension and reflects the overall state of stress at the volcano [e.g., *Cayol et al.*, 2000]. Geologic and geophysical evidence suggest that the east rift zone, and indeed Kīlauea's summit magma storage complex in general, has migrated south over time, presumably due to south flank instability [*Swanson et al.*, 1976]; the shallow magma pathway within the caldera may therefore represent

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an older rift system that has largely been abandoned except in the caldera. Rifting also provides an alternative model to explain the dynamic gravity increase observed in Kīlauea Caldera during 1975–2008 [*Johnson et al.*, 2010]; the continual expansion of magma reservoirs through rifting can supply the mass flux to produce the 450 μGal gravity increase without accompanying surface uplift.

[30] To explore the effect of rifting on Kīlauea's shallow magmatic system, we constructed a simple model of Kīlauea's shallow magma chamber (which is located beneath the east margin of Halema'uma'u Crater [*Cervelli and Miklius*, 2003]) by assuming a 1 km^3 inflating sphere at a depth of 1 km with an increasing spherical radius of 3 cm yr^{-1}. The modeled magma chamber volume is within previous estimates of 0.5 to 1.8 km^3 [*Johnson*, 1992; *Ohminato et al.*, 1998; *Poland et al.*, 2009], and the modeled radial increase is consistent with the rate of summit extension due to south flank motion [e.g., *Delaney et al.*, 1998]. The annual volume change ($\Delta V = 1.4 \times 10^5\ m^3$) is then treated as a Mogi source to determine mass flux. The gravity change, Δg, is described by adding the three equations (equations 3, 4, and 5) [*Lisowski*, 2007]:

$$g_o = G(\rho_o - \rho_c)\Delta V(z/R^3) \quad (3)$$

$$g_1 = G\rho_c 2(1-\nu)\Delta V(z/R^3) \quad (4)$$

$$g_2 = -G\rho_c(1-2\nu)\Delta V(z/R^3) \quad (5)$$

where G is the gravitational constant, ρ_o is the magma density, ρ_c is the density of the crust, ν (0.25) is the Poisson's ratio, z is the depth to the source, and R is the position of the source in Cartesian coordinates $(x^2+y^2+z^2)^{1/2}$. Assuming ρ_c is equal to 2300 kg m^{-3} (based on downhole geophysics [*Keller et al.*, 1979] and previous gravity surveys [*Kauahikaua et al.*, 2000]), the resulting increase in gravity over 33 years would be 265 μGal with no density contrast and 330 μGal with a density contrast of 700 kg m^{-3} (magma density of 3000 kg m^{-3}). In other words, 59% to 73% of the measured dynamic gravity signal can be accounted for in this simple first-order model; thus, with greater rates of rifting, it is possible that the entire dynamic gravity signal may be due to the rifting of summit magma storage areas. In a larger context, this modeled rifted volume (~$10^5\ m^3\ yr^{-1}$) represents approximately 0.1% of Kīlauea's average annual eruptive output since the start of the current east rift zone eruption in 1983 ($1.3 \times 10^8\ m^3$ [*Sutton et al.*, 2003]).

[31] In addition to explaining at least a portion of the dynamic gravity increase, rifting also provides a mechanism for the formation of approximately 0.7 km^3 of olivine cumulates suggested by our Bouguer gravity survey. Gravitational spreading due to seaward movement of the south flank, coupled with the usual magma supply to the volcano, provides a mechanism to build and expand magma reservoirs beneath the summit. Crystallization of these reservoirs would build large piles of olivine cumulates that would be left behind when magma reservoirs were evacuated and caldera collapsed ensued.

[32] The applicability of this simple conceptual model will need to be thoroughly tested through numerical modeling of gravitational spreading (rifting) and the continuation of geodetic (dynamic gravity and deformation) measurements. Numerical models that incorporate realistic geologic properties, such as crustal rheology, cumulate volume, and the effect of edifice buttressing, may best be able to describe the effects of rifting on the shallow magmatic system. Likewise, if dynamic gravity measurements are continued, the predicted long-term trend would be a gravity increase without accompanying surface uplift as long as summit rifting and south flank motion persists.

5. Conclusion

[33] The magnetic data identify two nontopographic anomalies within Kīlauea Caldera, corresponding to shallow structural features. The northernmost anomaly is likely due to either the long-lived prehistoric Observatory vent (470 to 625 B.P. [*Holcomb*, 1987; *Neal and Lockwood*, 2003]) or a buried lava lake [*Ellis*, 1825]. The second anomaly is probably related to a set of eruptive fissures from 1954, 1971, 1974, and 1982 that strike northeast from Halema'uma'u Crater. There may be older eruptive fissures, now obscured by more recent lava flows, which used the same structure, as it appears to be an important magma pathway within the caldera. These data expand the current knowledge of the structures within the caldera and provide a basis for further investigations which may be able to identify other buried fissure zones or long-sustained eruptive vents.

[34] Attempts to constrain the amount of void space in the summit region of Kīlauea have been made; however, no quantitative conclusion could be reached due to nonuniqueness in the interpretation of gravitational fields (void space can be masked by denser material above or below). The positive Bouguer gravity anomaly centered within Kīlauea Caldera can be modeled as a large, shallow (<2 km) intrusive complex of approximately 3.0 km^3 consisting of dense solidified intrusions, olivine cumulates, and shallow magma reservoirs offset to the northeast from Halema'uma'u Crater. Creation of such an intrusive body would be facilitated by rifting of the summit, which is known to be occurring from geodetic data [e.g., *Delaney et al.*, 1998]. The body we image at the center of the caldera may represent the cumulate body that remained after drainage of a previous magma reservoir prior to formation of the current caldera.

[35] **Acknowledgments.** This study was supported by a NSERC Discovery grant to G. Williams-Jones and a Kleinman grant to J. Zurek. Mahalo to Albert Eggers, Jim Kauahikaua, Hazel Rymer, Matthew Patrick, and Tim Orr for helpful discussions and to Rick Blakely, Micol Todesco, Nicolas Fournier, and Giovanna Berrino for their constructive reviews. This study would not have been possible without the significant contributions of Mike Poland and Dan Dzurisin. Also, thanks to the staff of HVO and Hawai'i Volcanoes National Park for their support.

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